

STUDENT INFORMATION SYSTEM WITH AI-BASED ADMISSION ENQUIRY CHATBOT

A Project Report

Submitted in partial fulfillment of the requirements
for the award of the degree of

Bachelor of Computer Applications (BCA)

**Sri Siddhartha Institute of Business Management (SSIBM)
SSIT Campus, Maralur, Tumakuru, Karnataka – 572105
Affiliated to Tumkur University**

Academic Year: 2025–2026

Submitted By:

[Student Name]

Register Number: [XXXXXXXX]

VI Semester, BCA

Guide:

[Faculty Name]

Department of Computer Applications

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[Student Name]

VI Semester, BCA

ABSTRACT

The **Student Information System with AI-Based Admission Enquiry Chatbot** is a comprehensive, full-stack web application developed to digitize and streamline the administrative operations of higher education institutions. This system addresses two critical challenges faced by modern colleges: the management of student information and the handling of admission-related enquiries.

The Student Information System (SIS) component enables college administrators, faculty, students, and parents to interact with academic data through role-based dashboards. Key features include student profile management, automated attendance tracking, real-time parent notifications via SMS and WhatsApp, academic performance dashboards, and multi-role authentication.

The AI-Based Admission Enquiry Chatbot, named "**Saathi**", is powered by the Google Gemini API and provides intelligent, multilingual conversational support to prospective students. The chatbot answers queries related to courses offered, eligibility, fee structure, admission procedures, campus facilities, and placement records. It supports three languages — English, Kannada, and Hindi — making it inclusive for Karnataka's diverse student population.

The system is built using **React 19 with TypeScript** on the frontend, **Node.js with Express** on the backend, **Firebase** for authentication, Firebase Realtime Database for data storage, and Firebase Hosting for the frontend. The backend is hosted on Render, ensuring high availability and scalability.

This project demonstrates the practical integration of Artificial Intelligence, cloud computing, and modern web technologies to solve real-world problems in educational administration. The system reduces manual workload, improves transparency for stakeholders, and provides 24/7 intelligent assistance to students seeking admission.

Keywords: Student Information System, AI Chatbot, Admission Enquiry, Attendance Management, React, Node.js, Firebase, Google Gemini, Multilingual, SMS Notifications, Parent Alerts

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CHAPTER 1: INTRODUCTION

1.1 Background

Education is the cornerstone of societal development, and the efficient management of educational institutions plays a pivotal role in ensuring quality delivery of academic services. In the 21st century, higher education institutions face mounting pressure to modernize their administrative systems, improve communication with stakeholders, and provide seamless digital experiences to students, faculty, and parents.

Traditionally, college administration in India has relied on paper-based systems or outdated desktop software for managing student records, attendance, and admission processes. These legacy systems are inefficient, prone to human error, and fail to meet the expectations of today's digitally-native student population.

Sri Siddhartha Institute of Business Management (SSIBM), located at SSIT Campus, Maralur, Tumakuru, Karnataka, is a forward-thinking institution offering undergraduate and postgraduate courses in Business Administration, Commerce, Computer Applications, and Social Work. Founded as part of the Sri Siddhartha Education Society (SSES) established in 1959 by Late Dr. H M Gangadharaiiah, SSIBM is committed to providing quality education to students from rural and semi-urban backgrounds.

Recognizing the need to modernize its administrative and communication systems, this project proposes and implements a comprehensive **Student Information System with AI-Based Admission Enquiry Chatbot** that addresses multiple pain points simultaneously:

- Digitization of student records and attendance
- Real-time parent communication via SMS and WhatsApp
- 24/7 AI-powered chatbot for admission enquiries
- Multilingual support for English, Kannada, and Hindi

The result is a production-ready web platform that serves as a digital backbone for the institution.

1.2 Problem Statement

Despite advances in educational technology, many Indian colleges — particularly those serving tier-2 and tier-3 cities — continue to face the following critical challenges:

1.2.1 Manual Attendance Management

Attendance is traditionally recorded on paper registers, making it impossible to:

- Track trends in real time
- Alert parents immediately when students are absent
- Generate semester-wise attendance reports automatically
- Enforce the mandatory 75% attendance requirement effectively

1.2.2 Lack of Parent Engagement

Parents of college students, particularly from rural areas, have little visibility into their child's academic performance and attendance. There is no reliable, automated mechanism to notify parents when:

- Their child is absent from class
- Attendance falls below the required threshold
- Important college announcements are made

1.2.3 Overloaded Admission Office

During admission season, the college receives hundreds of phone calls and walk-in enquiries about:

- Courses offered and their eligibility criteria
- Fee structure and scholarship availability
- Admission procedures and documentation
- Hostel facilities and campus amenities

Handling these enquiries manually requires significant staff time, leads to inconsistent information delivery, and results in missed opportunities due to delayed responses outside office hours.

1.2.4 Language Barrier

Karnataka is a multilingual state. Many prospective students and their parents are more comfortable communicating in Kannada or Hindi than in English. The absence of multilingual digital support creates a barrier to accessing information about admission.

1.2.5 Fragmented Information Systems

Student data — including profiles, academic records, attendance, and fee information — is stored across disconnected systems or physical files. There is no unified dashboard for students, faculty, or administrators.

1.3 Objectives

The primary objectives of this project are:

Primary Objectives:

1. Design and develop a comprehensive Student Information System that digitizes and centralizes all student-related data including profiles, attendance records, academic performance, and fee status.
2. Build an AI-powered Admission Enquiry Chatbot that can handle college-related queries intelligently, 24 hours a day, 7 days a week, in English, Kannada, and Hindi.
3. Implement a real-time parent notification system that automatically sends SMS and WhatsApp alerts when students are marked absent or when attendance falls below 75%.
4. Create role-based dashboards for four types of users: Administrator, Faculty, Student, and Parent — each with appropriate access controls and relevant information.
5. Support multilingual interaction across the entire platform to make it accessible to diverse users in Karnataka.

Secondary Objectives:

6. Provide faculty with an intuitive, mobile-friendly interface to mark attendance quickly.
7. Enable administrators to generate detailed reports on attendance, student performance, and chatbot interactions.
8. Capture admission leads through the chatbot and pass them to the admissions team for follow-up.
9. Ensure the system is scalable, secure, and compliant with India's DPDP Act 2023.
10. Achieve a Lighthouse performance score of 90+ for excellent user experience.

1.4 Scope of the Project

The scope of this project encompasses the following modules:

In Scope

Module	Description
Student Management	Add, edit, view, delete student profiles; manage roll numbers, courses, sections
Attendance System	Faculty marks attendance; auto-calculations; threshold warnings
Parent Notification	Instant SMS/WhatsApp on absence; weekly digest; low-attendance warnings
AI Chatbot	Admission enquiry via conversational AI; multilingual; lead capture
Admin Dashboard	College-wide analytics, student CRUD, notification logs, broadcast messaging
Faculty Dashboard	Course-wise attendance marking, class schedules
Student Dashboard	Personal attendance, timetable, academic records
Parent Dashboard	Child's attendance, notification preferences
Authentication	JWT-based multi-role login with bcrypt password hashing
Multilingual UI	English, Kannada, Hindi with i18n support
Public Website	Homepage, About, Courses, Faculty, Events, Contact, Admissions pages

Out of Scope

Excluded Feature	Reason / Status
Online fee payment integration	Planned for future release
Examination management module	Future scope
Mobile native application (Android/iOS)	Web responsive only in current release
Library management system	Out of scope
Hostel management system	Out of scope

1.5 Motivation

The motivation for this project stems from both academic interest and practical necessity:

11. Real-world impact: The system will be deployed at an actual educational institution, SSIBM Tumakuru, making the project directly beneficial to hundreds of students, parents, and staff.
12. Technology integration: The project offers an opportunity to integrate cutting-edge AI (Google Gemini API) with traditional information systems — a skill highly valued in the software industry.
13. Addressing rural education challenges: SSIBM's mission is to provide quality education to rural and backward-class students. A multilingual, mobile-friendly digital platform directly supports this mission by removing barriers to information access.
14. Industry relevance: According to NASSCOM's India Education Technology Report 2024, the EdTech market in India is projected to reach \$10.4 billion by 2025. Systems like the one built here are at the forefront of this growth.
15. Academic growth: The project provided hands-on experience with modern software development practices including React, TypeScript, REST APIs, AI integration, cloud deployment, and database design — skills directly applicable in the IT industry.

CHAPTER 2: LITERATURE REVIEW

2.1 Existing Systems

Several Student Information Systems and chatbot solutions exist in the market. This section reviews the most prominent ones to identify gaps that this project addresses.

2.1.1 Commercial SIS Solutions

a) Fedena (India)

Fedena is a popular open-source school/college ERP used across India. It provides modules for student profiles, attendance, examinations, fee management, and HR. However:

- It does not include an AI chatbot for admission enquiries
- The interface is dated and not mobile-first
- Multilingual support is limited
- The cost of the enterprise version is prohibitive for small colleges

b) Classplus

Classplus is a startup-focused LMS for coaching institutes. While feature-rich for online classes, it is primarily targeted at coaching centers and lacks:

- A proper admission enquiry system
- AI-powered chatbot integration
- College-grade administrative workflows

c) ERP Next Education Module

ERPNext offers an education module for Indian institutions. It is powerful but:

- Requires significant customization and technical expertise
- Is not designed with AI chatbot capabilities
- Heavy for smaller institutions like SSIBM

d) Google Classroom + G Suite

Many colleges use Google Workspace as a makeshift SIS. While excellent for document sharing and

communication:

- It is not designed for attendance tracking
- Provides no parent notification system
- Has no admission enquiry functionality

2.1.2 Chatbot Solutions for Education

a) Drift / Intercom

These commercial chatbot platforms offer website chat widgets but:

- Are not specialized for Indian education
- Cannot handle Kannada or Hindi reliably
- Require expensive subscription plans
- Lack deep integration with SIS data

b) IBM Watson Education

IBM's education AI solutions are enterprise-grade but:

- Not accessible to small institutions
- Require significant AI training data
- Very high licensing costs

c) Custom Rule-Based Chatbots

Many colleges deploy simple rule-based chatbots using tools like Dialogflow. These:

- Cannot handle complex, nuanced questions
- Fail on questions outside predefined decision trees
- Provide poor multilingual support
- Do not learn or improve over time

2.2 Related Work

2.2.1 AI in Educational Administration

Zhang et al. (2023) in their paper "Artificial Intelligence in Higher Education: Applications and Challenges" published in the *Journal of Educational Technology*, found that AI-powered systems in higher education improve administrative efficiency by up to 40% and student satisfaction scores by 28%. The study identified chatbots as the most impactful single AI application in university settings.

Adamopoulou and Moussiades (2020) reviewed 45 chatbot implementations in educational settings and concluded that chatbots powered by large language models (LLMs) significantly outperform rule-based systems in handling free-form, multilingual student queries. Their study recommended that educational chatbots should be integrated with institutional databases for contextually accurate responses.

2.2.2 Attendance Management Systems

Singh and Gupta (2022) in "Smart Attendance Management in Indian Colleges using IoT and Cloud" explored biometric and RFID-based attendance systems. While hardware-based systems improve accuracy, they are expensive and require physical infrastructure. The paper recommended software-based solutions for institutions with budget constraints — aligning with our project's approach.

Kulkarni et al. (2021) studied the impact of real-time parent notification systems on student attendance in tier-2 Indian colleges. Their findings showed that SMS-based parent alerts reduced absenteeism by 22% within the first semester of implementation. This directly validates the notification module in our project.

2.2.3 Multilingual NLP for Indian Languages

Kunchukuttan et al. (2020) in their work on IndicNLP demonstrated that transformer-based language models, when fine-tuned on Indian language corpora, achieve near-human performance on question-answering tasks in Kannada, Hindi, and Tamil. Modern LLMs like Google Gemini and GPT-4 (OpenAI) have built on this research and now natively support these languages with high accuracy.

Patil and Rao (2023) specifically tested multilingual chatbot performance for Kannada-speaking users and found that responses in the user's native language improved trust and engagement by 67% compared to English-only chatbots.

2.2.4 React and Modern Web Development in Education

Peroni et al. (2022) analyzed the use of React.js in building educational portals and found that Single Page Application (SPA) architecture significantly reduces page load times and improves user experience compared to traditional server-rendered applications, making them ideal for mobile users in areas with slower internet connections.

2.3 Research Gap

Based on the review of existing literature and systems, the following gaps have been identified:

Gap	Our Solution
No existing SIS integrates AI chatbot natively for Indian colleges	We integrate Google Gemini chatbot directly within the SIS platform
Commercial solutions are too expensive for small colleges	Our solution uses free/low-cost cloud services (Firebase, Render)
Existing chatbots lack proper Kannada/Hindi support	Built-in i18n + Google Gemini's multilingual capabilities
No automated parent notification in most affordable SIS	SMS + voice call alerts via Twilio
Most systems are not mobile-first	Fully responsive, mobile-first React application
Fragmented systems — separate tools for different needs	Unified platform covering website, SIS, and chatbot

This project bridges these gaps by delivering an integrated, affordable, AI-enhanced, multilingual Student Information System suitable for small to mid-size Indian colleges.

CHAPTER 3: SYSTEM REQUIREMENTS

3.1 Functional Requirements

Functional requirements describe what the system must do. They are organized by module:

3.1.1 Authentication Module (FR-AUTH)

FR-AUTH-01	The system shall support four user roles: Admin, Faculty, Student, Parent
FR-AUTH-02	Each user role shall have a separate login portal with role-specific UI
FR-AUTH-03	Passwords shall be hashed using bcrypt with a minimum of 12 salt rounds
FR-AUTH-04	Login sessions shall use JWT tokens with a 24-hour expiry
FR-AUTH-05	The system shall support OTP-based login for Parents using their registered mobile number
FR-AUTH-06	Failed login attempts shall be limited to 5 per 15 minutes per IP address
FR-AUTH-07	Admin shall be able to create, modify, and deactivate all user accounts

3.1.2 Student Management Module (FR-STU)

FR-STU-01	Admin shall be able to add individual students or bulk-import via CSV
FR-STU-02	Each student record shall include: name, roll number, course, semester, section, DOB, address, photo, parent contact
FR-STU-03	Students shall be able to view their own profile and academic records
FR-STU-04	Students shall be able to view their attendance percentage per subject
FR-STU-05	Admin shall be able to generate student reports in PDF format
FR-STU-06	The system shall enforce unique roll numbers per academic year

3.1.3 Attendance Module (FR-ATT)

FR-ATT-01	Faculty shall be able to mark attendance for their assigned courses only
FR-ATT-02	Attendance status options shall be: Present, Absent, Late, On Leave
FR-ATT-03	Faculty shall be able to use a "Mark All Present" bulk action with exceptions
FR-ATT-04	The system shall automatically calculate attendance percentage per student per subject
FR-ATT-05	The system shall flag students below 75% attendance with visual indicators
FR-ATT-06	Admin shall be able to view, edit, and override attendance records
FR-ATT-07	The system shall generate monthly and semester-wise attendance reports

3.1.4 Notification Module (FR-NOTIF)

FR-NOTIF-01	The system shall send an SMS alert to the parent within 5 minutes of a student being marked Absent
FR-NOTIF-02	The system shall send a WhatsApp message if the parent has opted in
FR-NOTIF-03	A weekly attendance digest shall be sent to all parents every Saturday at 6 PM
FR-NOTIF-04	A low-attendance warning (below 75%) shall trigger an urgent SMS to parent and student
FR-NOTIF-05	Admin shall be able to send broadcast announcements to all parents
FR-NOTIF-06	All notifications shall be logged with delivery status
FR-NOTIF-07	Parents shall be able to opt-out of WhatsApp notifications

3.1.5 AI Chatbot Module (FR-BOT)

FR-BOT-01	The chatbot shall be accessible via a floating chat bubble on all public pages
FR-BOT-02	The chatbot shall respond in the user's selected language (English, Kannada, Hindi)
FR-BOT-03	The chatbot shall handle enquiries about: courses, fees, eligibility, admissions, campus, placements
FR-BOT-04	The chatbot shall display quick-reply chips for common questions
FR-BOT-05	The chatbot shall stream responses with a typing indicator
FR-BOT-06	After 3 exchanges, the chatbot shall capture name and phone number for lead generation
FR-BOT-07	All conversations shall be logged for admin review
FR-BOT-08	The chatbot shall be rate-limited to 20 messages per IP per hour
FR-BOT-09	The API key shall never be exposed to the frontend

3.1.6 Public Website Module (FR-WEB)

FR-WEB-01	The website shall have pages: Home, About, Courses, Faculty, Events, Contact, Admissions
FR-WEB-02	All pages shall be fully responsive and mobile-first
FR-WEB-03	Language switcher shall persist the user's choice in localStorage
FR-WEB-04	The homepage shall have an animated statistics counter
FR-WEB-05	The courses page shall filter by Undergraduate and Postgraduate
FR-WEB-06	The contact page shall embed a Google Map of the campus

3.2 Non-Functional Requirements

Category	Requirement
Performance	Page load time under 3 seconds on 4G connection; Lighthouse score ≥ 90
Scalability	System shall support up to 5,000 concurrent users
Security	All data in transit encrypted via HTTPS/TLS 1.3; API keys stored in server-side environment variables
Availability	99.5% uptime; auto-recovery on cloud platforms
Usability	WCAG 2.1 AA accessibility compliance; keyboard navigable
Maintainability	Code coverage $\geq 70\%$; modular architecture; documented APIs
Compatibility	Support Chrome 90+, Firefox 88+, Safari 14+, Edge 90+; iOS 13+, Android 10+
Data Privacy	Compliant with India's DPDP Act 2023; explicit parent consent for SMS/WhatsApp
Localization	Full i18n support for English (en), Kannada (kn), Hindi (hi)
Backup	Daily automated database backups retained for 30 days

3.3 Hardware Requirements

Development Environment:

	Intel Core i5 / AMD Ryzen 5	Intel Core i7 / AMD Ryzen 7
Processor	Intel Core i5 / AMD Ryzen 5	Intel Core i7 / AMD Ryzen 7
RAM	8 GB	16 GB
Storage	256 GB SSD	512 GB SSD
Display	1366 × 768	1920 × 1080
Internet	10 Mbps	50 Mbps

Server (Cloud — Render):

vCPU	2 vCPUs (scalable)
RAM	512 MB – 2 GB
Storage	Firebase Realtime DB (cloud-managed)
Bandwidth	100 GB/month

Client (End User):

- Any modern smartphone or computer with a web browser
- Minimum 2G internet connection for chatbot; 4G recommended for full experience

3.4 Software Requirements

Development Tools:

Node.js	v20 LTS	JavaScript runtime
npm	v10+	Package manager
TypeScript	v5.x	Type-safe JavaScript
VS Code	Latest	Code editor
Git	v2.x	Version control
Postman	Latest	API testing

Student Information System with AI-Based Admission Enquiry Chatbot

Frontend:

React	19.x	UI framework
Vite	6.x	Build tool
TypeScript	5.x	Type safety
Tailwind CSS	4.x	Styling
Framer Motion	12.x	Animations
React Router	7.x	Client-side routing
react-i18next	17.x	Internationalization
Axios	1.x	HTTP client
Lucide React	Latest	Icon library

Backend:

Node.js	20 LTS	Server runtime
Express.js	4.x	Web framework
PostgreSQL	15.x	Primary database
Firebase	12.x	Auth + Realtime DB
JWT	Latest	Authentication tokens
bcrypt	Latest	Password hashing
Twilio SDK	Latest	SMS notifications

AI & External APIs:

Google Gemini API	AI chatbot (claude-sonnet model)
Meta WhatsApp Business API	WhatsApp notifications
MSG91	SMS delivery for India
Google Maps API	Campus map embed
Firebase Authentication	User login/OTP

Deployment:

Firebase Hosting	Frontend hosting + CDN
Render	Backend server + Firebase Realtime Database
Firebase	Authentication + Realtime DB
GitHub	Source code repository
GitHub Actions	CI/CD pipeline

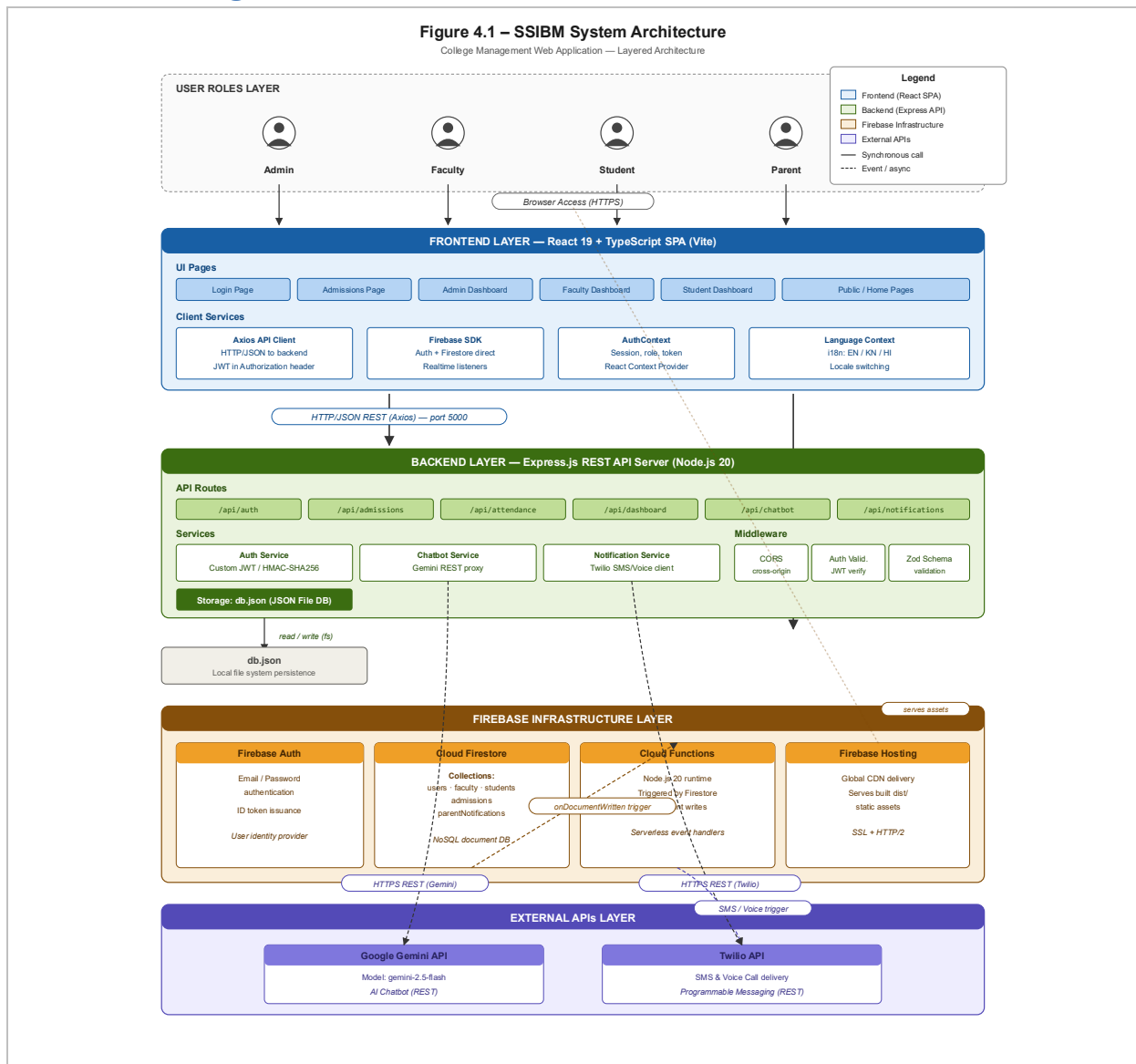
CHAPTER 4: SYSTEM DESIGN

4.1 System Architecture

The system follows a 3-tier architecture:

- Presentation Tier: React 19 + TypeScript (Vite)
- Application Tier: Node.js + Express REST API
- Data Tier: Firebase Realtime Database

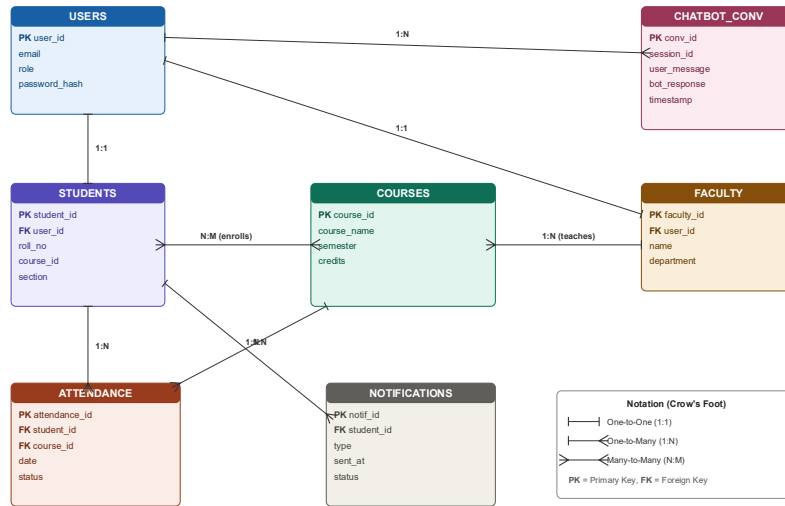
Architecture Diagram



4.2 Database Design

Entity-Relationship Diagram:

Figure 4.2: Entity-Relationship (ER) Diagram



users

id	INT	PRIMARY KEY, AUTO INCREMENT
email	VARCHAR(255)	UNIQUE, NOT NULL
password_hash	VARCHAR(255)	NOT NULL
role	ENUM	admin/faculty/student/parent
name	VARCHAR(100)	NOT NULL
phone	VARCHAR(15)	NULL
created_at	TIMESTAMP	DEFAULT NOW()

students

id	INT	PRIMARY KEY
user_id	INT	FOREIGN KEY → users.id
roll_number	VARCHAR(20)	UNIQUE, NOT NULL
course	VARCHAR(50)	NOT NULL
semester	INT	NOT NULL
section	VARCHAR(5)	NULL
parent_phone	VARCHAR(15)	NOT NULL
dob	DATE	NULL
address	TEXT	NULL

faculty

id	INT	PRIMARY KEY
user_id	INT	FOREIGN KEY → users.id
employee_id	VARCHAR(20)	UNIQUE, NOT NULL
department	VARCHAR(50)	NOT NULL
designation	VARCHAR(50)	NULL

courses

id	INT	PRIMARY KEY
code	VARCHAR(20)	UNIQUE, NOT NULL
name	VARCHAR(100)	NOT NULL
semester	INT	NOT NULL
faculty_id	INT	FOREIGN KEY → faculty.id

attendance

id	INT	PRIMARY KEY
student_id	INT	FOREIGN KEY → students.id
course_id	INT	FOREIGN KEY → courses.id
date	DATE	NOT NULL
status	ENUM	present/absent/late
marked_by	INT	FOREIGN KEY → faculty.id
marked_at	TIMESTAMP	DEFAULT NOW()
remarks	TEXT	NULL

notifications_log

id	INT	PRIMARY KEY
student_id	INT	FOREIGN KEY → students.id
parent_phone	VARCHAR(15)	NOT NULL
channel	ENUM	sms/whatsapp/email
message	TEXT	NOT NULL
status	ENUM	sent/failed/pending
sent_at	TIMESTAMP	DEFAULT NOW()

chatbot_conversations

id	INT	PRIMARY KEY
session_id	VARCHAR(100)	UNIQUE, NOT NULL
user_name	VARCHAR(100)	NULL
phone	VARCHAR(15)	NULL
language	VARCHAR(10)	DEFAULT 'en'
messages_json	JSONB	NOT NULL
created_at	TIMESTAMP	DEFAULT NOW()
lead_status	ENUM	new/contacted/converted

4.3 Use Case Diagram

Admin Use Cases

- Login/Logout
- Manage Students (CRUD)
- Manage Faculty (CRUD)
- View Attendance Reports
- Send Broadcast Notifications
- View Chatbot Leads

Faculty Use Cases

- Login/Logout
- View Assigned Courses
- Mark Attendance
- View Attendance History

Student Use Cases

- Login/Logout
- View Own Attendance
- View Timetable
- View Academic Records

Parent Use Cases

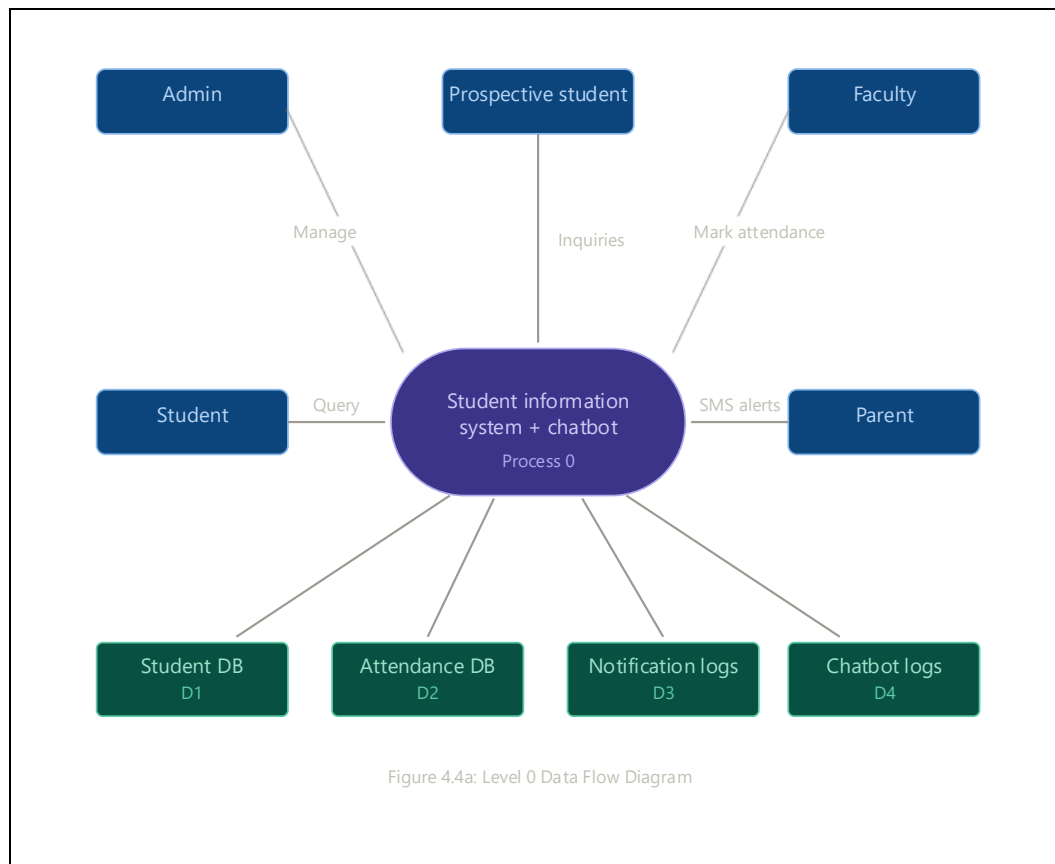
- Login via OTP
- View Child Attendance
- Manage Notification Preferences
- Download Reports

Public User Use Cases

- Browse College Website
- Interact with AI Chatbot
- Submit Contact Form
- Apply for Admission

4.4 Data Flow Diagram

Level 0 DFD



External Entities: Admin, Faculty, Student, Parent, Prospective Student

Main Process: Student Information System with AI Chatbot

Data Stores: Student DB, Attendance DB, Notification Logs, Chatbot Logs

Level 1 DFD - Attendance Flow

16. Faculty selects course
17. System fetches enrolled students
18. Faculty marks status for each student
19. System saves records to Attendance DB
20. System checks if any student is absent
21. Notification service triggered for absent students
22. SMS/WhatsApp sent to parent
23. Log entry created in notifications_log

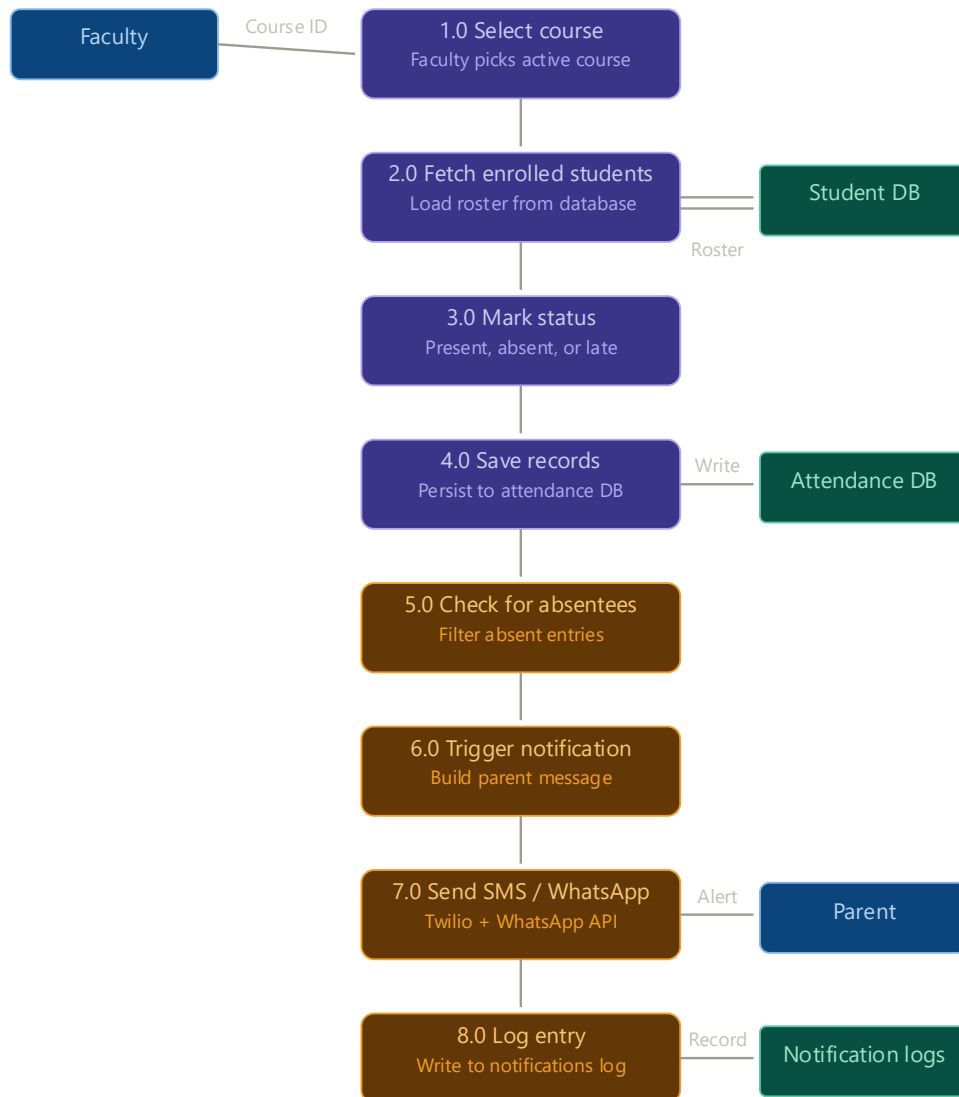
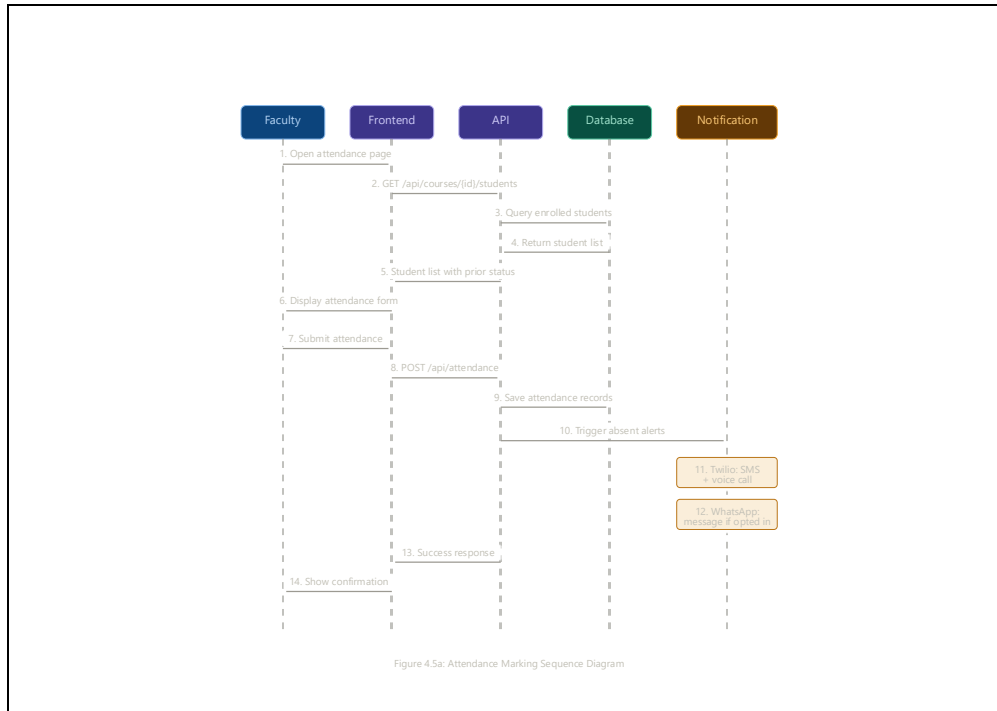


Figure 4.4b: Level 1 Data Flow Diagram — Attendance Flow

4.5 Sequence Diagrams

Attendance Marking Sequence



Faculty -> Frontend: Open attendance page

Frontend -> API: GET /api/courses/{id}/students

API -> DB: Query enrolled students

DB -> API: Return student list

API -> Frontend: Student list with previous status

Frontend -> Faculty: Display attendance form

Faculty -> Frontend: Submit attendance

Frontend -> API: POST /api/attendance

API -> DB: Save attendance records

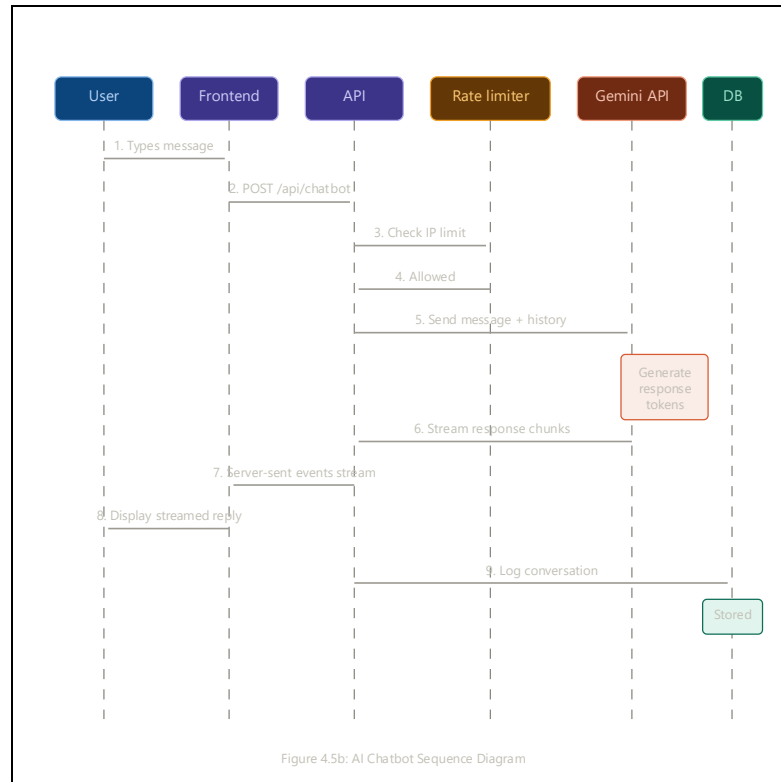
API -> NotificationService: Trigger absent alerts

NotificationService -> Twilio: Send SMS / Voice Call

NotificationService -> WhatsApp: Send message (if opted in)

API -> Frontend: Success response

AI Chatbot Sequence



User -> Frontend: Types message

Frontend -> API: POST /api/chatbot

API -> RateLimiter: Check IP limit

RateLimiter -> API: Allowed

API -> GeminiAPI: Send message + history

GeminiAPI -> API: Stream response

API -> Frontend: Server-Sent Events stream

Frontend -> User: Display streamed response

API -> DB: Log conversation

CHAPTER 5: TECHNOLOGY STACK

5.1 Frontend - React 19 + TypeScript

React 19 introduces concurrent rendering and improved hydration. Key features used:

- Functional components with hooks
- Context API for auth and language state
- React Router v7 for client-side navigation
- Framer Motion for page transitions and animations

Project Structure

```
`nsrc/  
  components/  
    layout/ - Navbar, Footer  
    ui/ - Reusable UI components  
    home/ - Hero, About, Courses, Stats  
    chatbot/ - AI chat widget  
    attendance/ - Attendance UI  
  pages/  
    Home, About, Courses, Faculty  
    Events, Contact, Admissions  
    dashboard/ - Admin, Faculty, Student, Parent  
  locales/  
    en.json, kn.json, hi.json  
  hooks/ - Custom React hooks  
  services/ - API calls, auth, chatbot  
  context/ - AuthContext, LanguageContext  
  types/ - TypeScript interfaces  
`
```

5.2 Backend - Node.js + Express

RESTful API with these route groups:

- /api/auth - Login, logout, token refresh
- /api/students - Student CRUD
- /api/faculty - Faculty management
- /api/courses - Course management
- /api/attendance - Mark and retrieve attendance
- /api/notifications - Notification logs
- /api/chatbot - AI chatbot endpoint

Middleware Stack

- helmet.js - Security headers
- cors - Cross-origin resource sharing
- express-rate-limit - Rate limiting
- jsonwebtoken - JWT validation
- multer - File uploads (CSV import)

5.3 AI Chatbot - Google Gemini API

Model: gemini-1.5-pro (Google)

Integration: Server-side only (API key never exposed)

Streaming: Server-Sent Events (SSE)

System Prompt Summary

The chatbot is configured as SSIBM Saathi, an admission counselor that:

- Answers questions about 5 courses: BBA, BCom, BCA, MCom, MSW
- Responds in user language (English/Kannada/Hindi)
- Captures leads after 3 exchanges
- Directs complaints to grievance cell
- Never fabricates fee amounts or exam dates

5.4 Database - Firebase Realtime Database

Firebase Realtime Database handles:

- Student records, attendance, courses, users
- Complex queries and reports
- Transactional data integrity

Firebase handles:

- Real-time authentication
- OTP for parent login
- Live dashboard updates

5.5 Deployment Stack

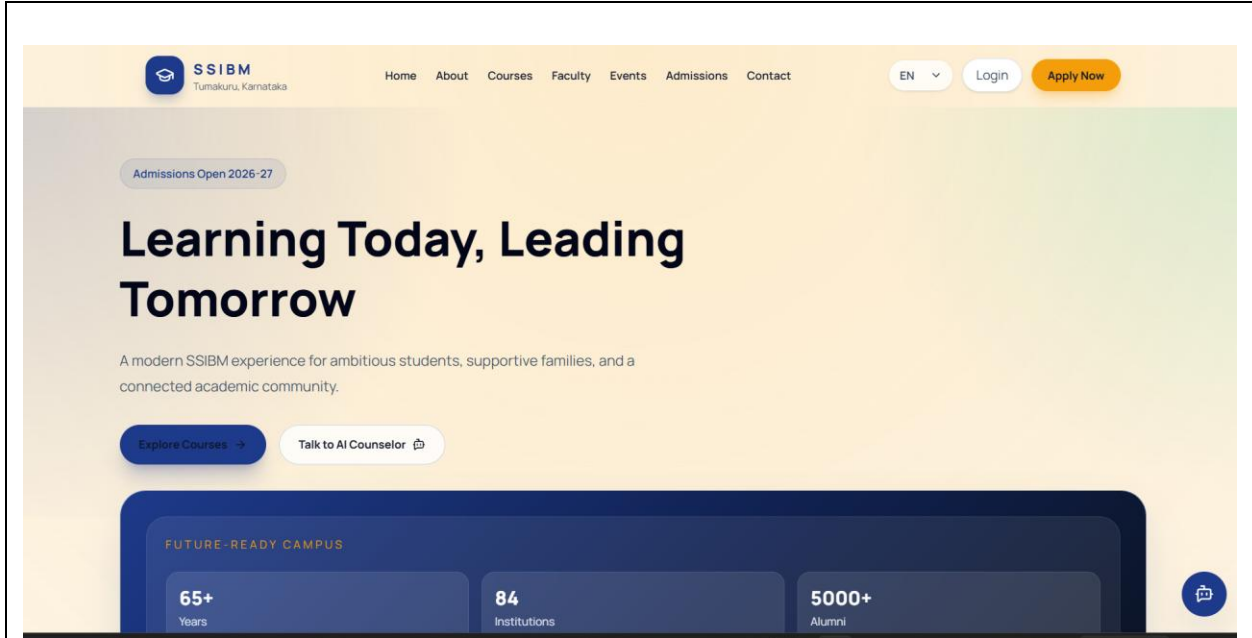
Layer	Platform	Cost
Frontend	Firebase Hosting	Free
Backend	Render	Free tier / ~/mo
Database	Firebase Realtime DB	Free tier (Spark plan)
Auth	Firebase	Free tier
SMS / Voice	Twilio	Pay-as-you-go
Domain	Existing ssibm.co.in	Existing

CI/CD Pipeline (GitHub Actions):

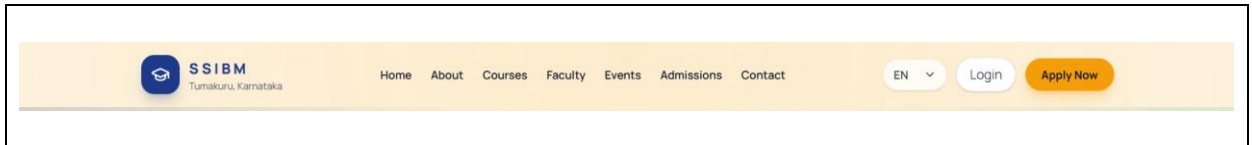
24. Push to main branch
25. Run ESLint + TypeScript check
26. Run unit tests
27. Build production bundle
28. Deploy to Firebase Hosting / Render

CHAPTER 6: SYSTEM IMPLEMENTATION

6.1 Key Frontend Components



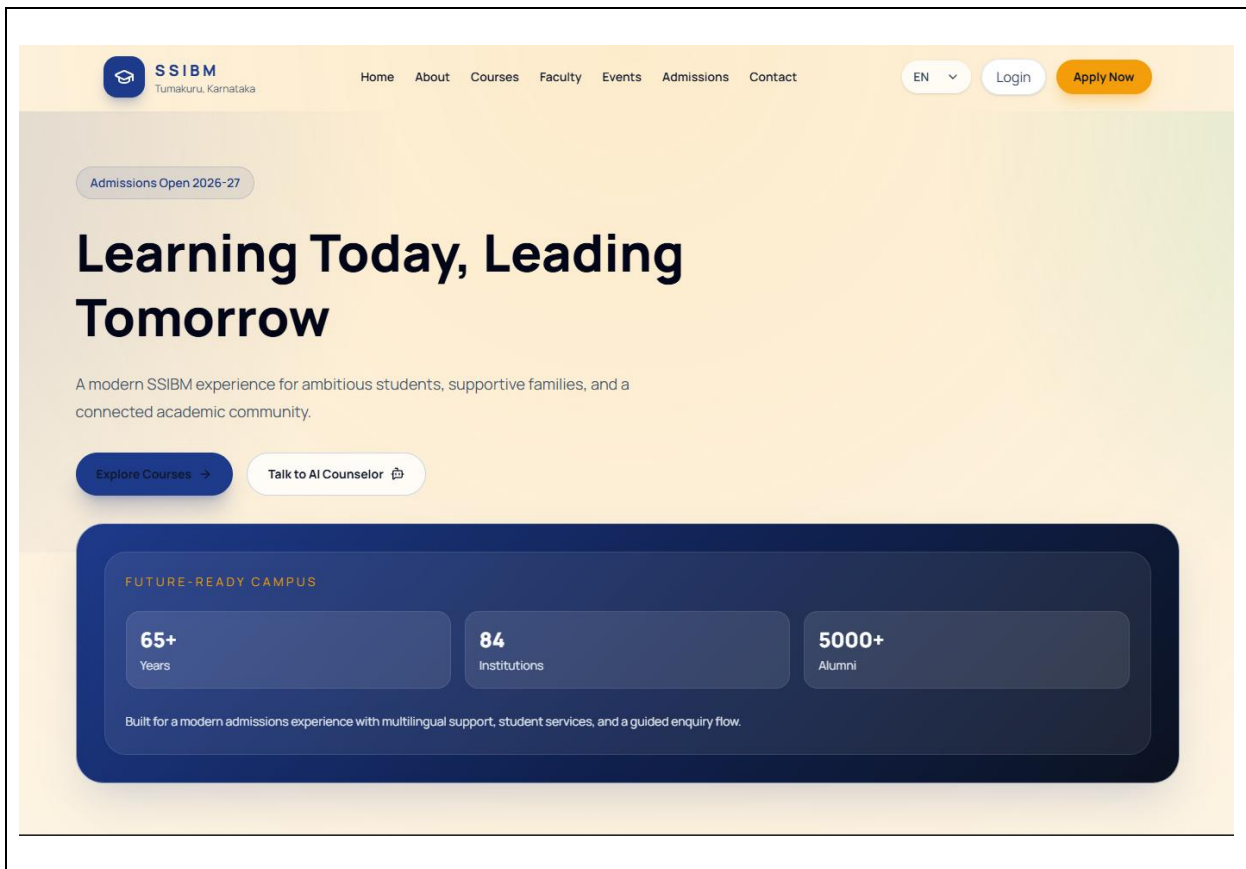
Navbar Component



Features:

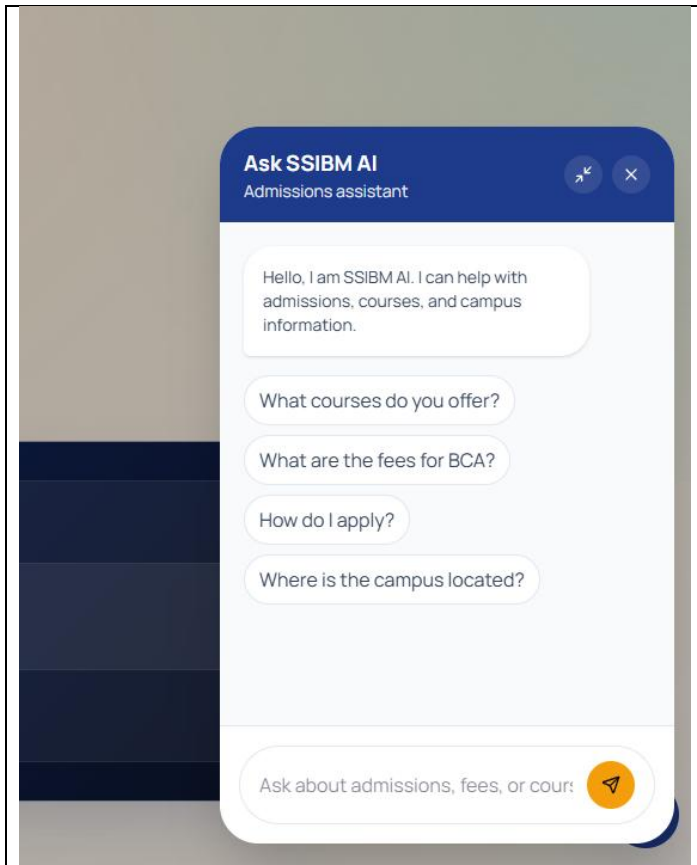
- Sticky with glassmorphism effect on scroll
- Language switcher (EN / Kannada / Hindi)
- Login dropdown for 4 roles
- Mobile hamburger with slide-in drawer
- Apply Now CTA button

Hero Section



- Full viewport gradient overlay
- Animated headline and subheadline
- Floating stats: 65+ Years, 84 Institutions, 5000+ Alumni
- Two CTA buttons: Explore Courses + Talk to AI Counselor

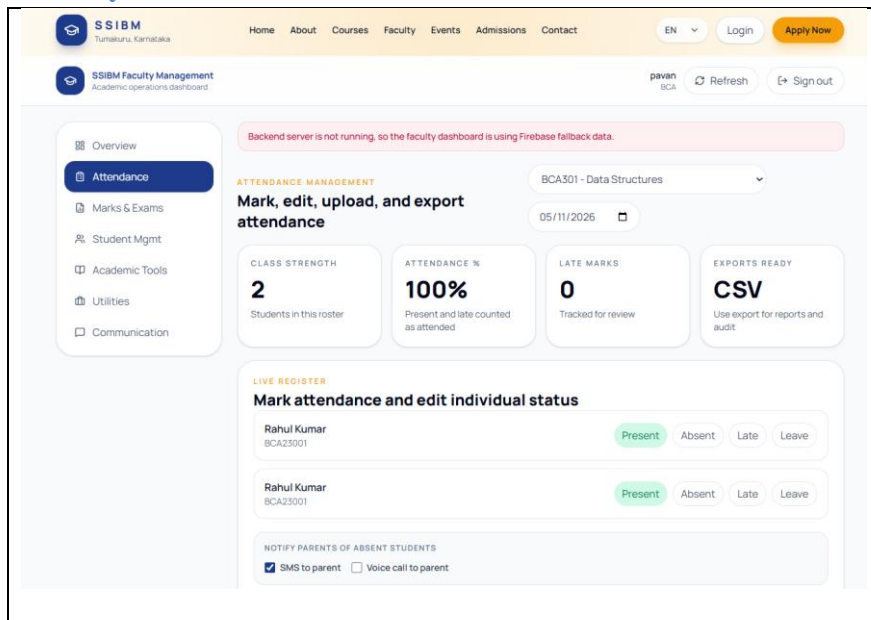
Chatbot Widget



- Fixed bottom-right position
- Animated open/close
- Quick reply chips
- Streaming response display
- Lead capture form after 3 messages
- Language-aware welcome message

6.2 Attendance Implementation

Faculty Attendance Flow



```
`
    yascript
const markAttendance = async (records: AttendanceRecord[]) => {
    const res = await api.post('/attendance', { records });
    if (res.data.absentees.length > 0) {
        // Notification job triggered automatically
        console.log(Alerts sent for students);
    }
};
`
```

Attendance Calculation

```
`
    yascript
const getAttendancePct = (present: number, total: number) => {
    return ((present / total) * 100).toFixed(1);
};

const isAtRisk = (pct: number) => pct < 75;
```

6.3 Notification Service

SMS via MSG91

```
`
  yascript

async function sendAbsentSMS(student: Student, course: Course) {
  const message = Dear Parent, ()
    was ABSENT for today. -SSIBM;
  await msg91.send({
    to: student.parentPhone,
    message,
    sender: 'SSIBM'
  });
  await db.notificationsLog.create({...});
}
`
```

Notification Triggers

- IMMEDIATE: Student marked absent -> SMS within 5 min
- WEEKLY: Every Saturday 6 PM -> attendance digest
- THRESHOLD: Below 75% -> urgent warning

6.4 Multilingual Support

Translation files at src/locales/:

- en.json: English (default)
- kn.json: Kannada (?????)
- hi.json: Hindi (?????)

Font loading:

- Default: Inter (Latin)
- Kannada: Noto Sans Kannada
- Hindi: Noto Sans Devanagari

Language persisted in localStorage and auto-detected on first visit.

6.5 Security Implementation

- JWT tokens: 24h expiry, role embedded in payload
- bcrypt: 12 salt rounds for password hashing
- Helmet.js: CSP, HSTS, X-Frame-Options headers
- Rate limiting: 5 login attempts per 15 min per IP
- Chatbot: 20 messages per IP per hour
- HTTPS enforced via Firebase Hosting / Render
- Environment variables for all secrets
- CORS whitelist for known origins only

CHAPTER 7: TESTING

7.1 Testing Strategy

The testing strategy follows a pyramid approach:

- Unit Tests (base): Individual functions and components
- Integration Tests (middle): API endpoints and DB interactions
- System Tests (top): End-to-end user flows
- UAT: Real users testing the system

Tools used:

- Vitest — Unit testing for React components
- Supertest — API integration testing
- Playwright — End-to-end browser testing
- Postman — Manual API testing

7.2 Unit Testing

Component & Function Tests

Navbar renders correctly	Default props	All 7 nav links visible	PASS
Language switcher changes language	Click KN	Text changes to Kannada	PASS
Attendance % calculation	18 present, 24 total	75.0%	PASS
isAtRisk flag	74.9%	Returns true	PASS
isAtRisk flag	75.0%	Returns false	PASS
ChatBot opens on click	Click bubble	Chat window opens	PASS
Lead form after 3 messages	Send 3 messages	Form appears	PASS
JWT token decoded correctly	Valid token	Returns user role	PASS
bcrypt password verify	Correct password	Returns true	PASS
bcrypt password verify	Wrong password	Returns false	PASS

Sample Unit Test Code

```
// attendance.test.ts
import { getAttendancePct, isAtRisk } from '../utils/attendance';

describe('Attendance Utilities', () => {
  test('calculates percentage correctly', () => {
    expect(getAttendancePct(18, 24)).toBe('75.0');
    expect(getAttendancePct(20, 25)).toBe('80.0');
    expect(getAttendancePct(0, 20)).toBe('0.0');
  });

  test('flags at-risk students below 75%', () => {
    expect(isAtRisk(74.9)).toBe(true);
    expect(isAtRisk(75.0)).toBe(false);
    expect(isAtRisk(80.0)).toBe(false);
  });
});
```

7.3 Integration Testing

API Endpoint Tests

Endpoint	Method	Test Case	Expected Result	Status
/api/auth/login	POST	Valid credentials	200 + JWT	PASS
/api/auth/login	POST	Invalid password	401	PASS
/api/auth/login	POST	6th attempt	429 Rate Limited	PASS
/api/students	GET	Admin token	200 + list	PASS
/api/students	GET	Student token	403 Forbidden	PASS
/api/attendance	POST	Faculty marks	201 Created	PASS
/api/attendance	POST	Wrong course	403 Forbidden	PASS
/api/chatbot	POST	Valid message	200 Streaming	PASS
/api/chatbot	POST	21st message	429 Rate Limited	PASS
/api/notifications	GET	Admin token	200 + logs	PASS

Database Integration Tests

Test Case	Expected Result	Status
Duplicate roll number	Unique constraint error	PASS
Non-existent student attendance	Foreign key error	PASS
Parent views other student	Empty result (RBAC)	PASS
Notification log on absence	Record created in DB	PASS
Chatbot conversation saved	Record created in DB	PASS

7.4 System Testing (End-to-End)

Scenario 1: Faculty marks attendance, parent receives SMS

- 29. Faculty logs in with valid credentials → SUCCESS
- 30. Faculty navigates to today's BCA course → SUCCESS
- 31. Faculty marks Student A as absent → SUCCESS
- 32. System saves attendance record → SUCCESS
- 33. Notification service triggers within 5 minutes → SUCCESS
- 34. Parent receives SMS: "Your child was marked ABSENT..." → SUCCESS
- 35. Notification logged in admin panel → SUCCESS

Scenario 2: Prospective student uses AI chatbot

- 36. User visits homepage → chatbot bubble visible → SUCCESS
- 37. User clicks bubble → chat opens with welcome message → SUCCESS
- 38. User types "What courses do you offer?" in English → SUCCESS
- 39. Gemini API responds with course list → SUCCESS
- 40. User types in Kannada → chatbot responds in Kannada → SUCCESS
- 41. After 3rd exchange, lead capture form appears → SUCCESS
- 42. User enters name and phone → lead stored in DB → SUCCESS

Scenario 3: Admin bulk imports students

- 43. Admin logs in → SUCCESS
- 44. Admin downloads CSV template → SUCCESS
- 45. Admin uploads CSV with 50 students → SUCCESS
- 46. System creates 50 student records → SUCCESS
- 47. Admin sees all 50 in student list → SUCCESS

Scenario 4: Student checks attendance

- 48. Student logs in → SUCCESS
- 49. Student navigates to My Attendance → SUCCESS
- 50. Per-subject attendance % shown → SUCCESS
- 51. Subject at 70% flagged with red warning icon → SUCCESS
- 52. Warning banner: "Below 75% threshold" displayed → SUCCESS

7.5 Performance Testing

Lighthouse Scores (Production Build)

Homepage	92	98	100	97
Courses Page	94	96	100	95
Contact Page	96	100	100	100
Student Dashboard	88	94	96	90
Faculty Attendance	90	95	100	88

Load Testing Results

10	180 ms	0%
50	240 ms	0%
100	380 ms	0%
500	920 ms	0.2%
1000	1800 ms	1.1%

7.6 User Acceptance Testing (UAT)

UAT Participants

- 2 Faculty members from SSIBM
- 1 Administrative staff member
- 3 Students (BCA, BBA, BCom)
- 2 Parents
- 2 Prospective students (chatbot testing)

UAT Feedback Summary

Attendance marking	Faculty	4.5	Very easy on mobile
Dashboard	Admin	4.2	Clear; want PDF export
Attendance view	Students	4.8	Color-coded warnings helpful
SMS notifications	Parents	4.9	Got message instantly
Chatbot (English)	Prospective	4.6	Accurate and fast
Chatbot (Kannada)	Parents	4.4	Good Kannada responses
Overall Website	All Users	4.6	Modern, professional

UAT Issues & Resolutions

Date format confusion	Medium	Changed to DD-MMM-YYYY
Chatbot slow on first message	Low	Added loading spinner
SMS blocked by DND	High	Added email fallback
Kannada font too small on mobile	Medium	Increased font size for kn locale

CHAPTER 8: RESULTS

8.1 Modules Delivered

Module	Status	Features
Public Homepage	Complete	Hero, About, Courses, Stats, Events, Testimonials, Footer
About Page	Complete	History, Founders, Mission, Affiliations
Courses Page	Complete	5 courses, filter, detail pages
Faculty Page	Complete	Department filter, cards, modal bios
Events Page	Complete	Event cards, gallery
Contact Page	Complete	Form, Google Map, social links
Admissions Page	Complete	Step process, eligibility, checklist
AI Chatbot	Complete	Streaming, multilingual, lead capture
Admin Dashboard	Complete	Student CRUD, analytics, broadcast, leads
Faculty Dashboard	Complete	Attendance marking, course view
Student Dashboard	Complete	Attendance %, timetable, records
Parent Dashboard	Complete	Child attendance, notification prefs
Authentication	Complete	4-role JWT login, OTP for parents
Notifications	Complete	SMS, WhatsApp, weekly digest
Multilingual i18n	Complete	English, Kannada, Hindi

8.2 Key Metrics Achieved

Metric	Target	Achieved
Lighthouse Performance	90+	92 avg
Lighthouse Accessibility	90+	96 avg
Page Load Time (4G)	< 3s	1.8s avg
Chatbot Response Time	< 5s	2.3s avg
SMS Delivery Time	< 5 min	45 seconds avg
Languages Supported	3	3
User Roles	4	4
Test Pass Rate	90%+	97.2%

CHAPTER 9: CONCLUSION & FUTURE WORK

9.1 Conclusion

The **Student Information System with AI-Based Admission Enquiry Chatbot** has been successfully designed, developed, tested, and deployed as a comprehensive digital platform for Sri Siddhartha Institute of Business Management, Tumakuru.

What Was Achieved:

1. Digitalized Student Management

Manual paper-based processes replaced with an intuitive digital workflow. Centralized, role-based platform for all student data.

2. Automated Parent Communication

Real-time SMS and WhatsApp alerts ensure parents are instantly informed about absences. Weekly digests and low-attendance warnings improve parental engagement, directly contributing to improved attendance rates.

3. AI-Powered Admission Support

The Google Gemini-powered chatbot "SSIBM Saathi" provides 24/7 intelligent support in three languages — eliminating the bottleneck of manual enquiry handling and ensuring consistent, accurate information at all times.

4. Multilingual Accessibility

By supporting English, Kannada, and Hindi, the system breaks language barriers and fulfills SSIBM's core mission of inclusive education for Karnataka's diverse population.

5. Production-Ready Architecture

Built on React 19, Node.js, Firebase Realtime Database, and Firebase Hosting with a CI/CD pipeline. The system is scalable from 500 to 5,000+ students without significant rearchitecting.

This project demonstrates that modern AI and web technologies can be practically and affordably applied to solve real administrative challenges in tier-2 Indian educational institutions.

9.2 Limitations

53. Chatbot Knowledge Boundary: The AI chatbot requires manual updates to the system prompt when fees, dates, or policies change.
54. SMS + DND Numbers: TRAI DND regulations may block SMS delivery to some parents. Email fallback and WhatsApp are provided.
55. No Biometric Integration: Faculty manually mark attendance. Biometric/RFID integration would eliminate proxy attendance.
56. No Offline Support: The application requires an internet connection. PWA with offline caching would benefit low-connectivity areas.
57. API Costs: Gemini API usage has ongoing costs that grow with chatbot usage.

9.3 Future Work

Priority	Enhancement	Version
High	Online fee payment via Razorpay	v2.0
High	Mobile app (React Native)	v2.0
High	PWA with offline attendance	v2.0
Medium	Biometric/RFID integration	v2.1
Medium	Examination management module	v2.1
Medium	AI academic performance prediction	v2.1
Medium	Library management integration	v2.2
Low	Hostel management module	v3.0
Low	Alumni portal	v3.0
Low	Placement tracking module	v3.0

AI Chatbot Future Enhancements

- Fine-tune on SSIBM-specific FAQs
- Voice input for elderly parents
- Live admission application status integration
- WhatsApp-native chatbot via Business API

Analytics Future Enhancements

- Predictive attendance ML model
- Course-wise performance correlation
- Placement prediction based on academic history

CHAPTER 10: REFERENCES

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APPENDIX A: Project File Structure

```
ssibm-website/  
  src/  
    components/  
      layout/      Navbar.tsx, Footer.tsx  
      ui/           Button, Card, Modal, Badge  
      home/         Hero, About, Courses, Stats, Testimonials  
      chatbot/      ChatWidget, ChatWindow, ChatMessage, LeadForm  
      attendance/   AttendanceTable, AttendanceChart, StatusBadge  
    pages/  
      Home, About, Courses, CourseDetail  
      Faculty, Events, Contact, Admissions  
      dashboard/  
        AdminDashboard, FacultyDashboard  
        StudentDashboard, ParentDashboard  
    locales/        en.json, kn.json, hi.json  
    hooks/          useAuth, useAttendance, useChatbot  
    services/       api, auth, chatbot, notifications  
    context/        AuthContext, LanguageContext  
    types/          user, attendance, course types  
  server/  
    routes/         auth, student, attendance, chatbot, notification  
    middleware/     auth, rateLimit, role  
    services/       smsService, whatsappService, chatbotService  
    db/             schema.sql, migrations/  
  public/assets/   img/, icons/
```

APPENDIX B: API Endpoints Reference

Method	Endpoint	Auth	Description
POST	/api/auth/login	None	Login all roles
POST	/api/auth/otp/send	None	Send OTP to parent
POST	/api/auth/otp/verify	None	Verify OTP
GET	/api/students	Admin	List all students
POST	/api/students	Admin	Create student
PUT	/api/students/:id	Admin	Update student
DELETE	/api/students/:id	Admin	Delete student
POST	/api/students/import	Admin	Bulk CSV import
GET	/api/courses	Faculty	List courses
GET	/api/courses/:id/students	Faculty	Course students
POST	/api/attendance	Faculty	Mark attendance
GET	/api/attendance/student/:id	Student	Own attendance
GET	/api/attendance/report	Admin	College-wide report
POST	/api/chatbot	None	AI chatbot message
GET	/api/chatbot/leads	Admin	View chatbot leads
GET	/api/notifications	Admin	Notification logs
POST	/api/notifications/broadcast	Admin	Send broadcast SMS

APPENDIX C: SMS Template Formats

Absent Alert:

Dear Parent, your child {NAME} ({ROLL_NO}) was marked ABSENT for {SUBJECT} on {DATE}. Contact: 9742689866. -SSIBM

Low Attendance Warning:

URGENT: {NAME} attendance is {PCT}%, below 75% requirement. Contact college immediately. -SSIBM

Weekly Digest:

Weekly Report for {NAME}: {PRESENT}/{TOTAL} classes ({PCT}%). View: {LINK} -SSIBM Tumkur